CIS 6270 Exam 1

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October 8, 2026

Please be certain to abide by the following rules for this exam:

- Please write your name on each page. You have 90 minutes.
- This exam is to be completed individually with no conferring with any classmates. You may ask questions individually with the instructor and teaching assistants. Do not consult other students within the class during or after submission about any content, questions, or answers for this exam until grades are returned.
- Calculators are not permitted. Work in the space provided, and box your final answer for every part. We will grade the boxed answer alone and read nothing else, so your scratch work costs you nothing.
- **Every part asks for exactly one boxed answer**, either a single equation or a single option number, so box one thing and one thing only in each part.
- Each labeled part is worth 0.5 points and receives either 0 or 0.5 points. There is no partial credit. Any form that is algebraically equivalent to the correct answer will earn full credit, so leave your answer in whatever form you reach it.
- You are only allowed a one page cheat sheet (8.5in $\times$ 11in, both sides can be used), and you may not use any devices (calculator, phone, etc.). You must submit the cheat sheet at the end of the exam, with your name signed on it.

The University of Pennsylvania is a community dedicated to scholarship, leadership, and service and to the principles of honesty, fairness, respect, and accountability. I acknowledge and agree to abide by my obligations under the Academic Integrity Policy. I will complete this exam in an honest, fair and trustworthy manner; my submitted answers will be entirely my own work and I will neither give nor receive any unauthorized assistance.

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This exam is worth 15 points (15% of your final grade).

Designing microRNAs with Adjustable Diffusion

You run a lab that designs microRNAs, which are RNA sequences about twenty-two nucleotides long that silence a gene by pairing with the untranslated region of that gene's mRNA transcript. As you know, no continuous sampler can move through sequences directly (that's for discrete samplers, which we will learn next), so you need to work in the embedding space of a frozen RNA language model, RNA-FM. Each microRNA becomes a tensor of RNA-FM embeddings, you generate a new tensor there, and the language-model head will decode it back to nucleotides. The state is a point in $\mathbb{R}^d$.

As you know by now, two generators are commonly available for continuous state space: flow matching, which trains a velocity field and generates by solving an ODE, and score-based diffusion, which trains a score and generates through reverse-time dynamics. In this exam, you will combine both fields into one sampler whose noise level you can turn up or down at generation time, while the distribution of latents at every intermediate time stays the one you chose. Adding Brownian noise changes the density evolution, so preserving that path requires a matching drift correction. You will derive it, prove agreement with the prescribed continuity equation, and then measure how finite steps, learned fields, and guidance affect distribution error and computational cost.

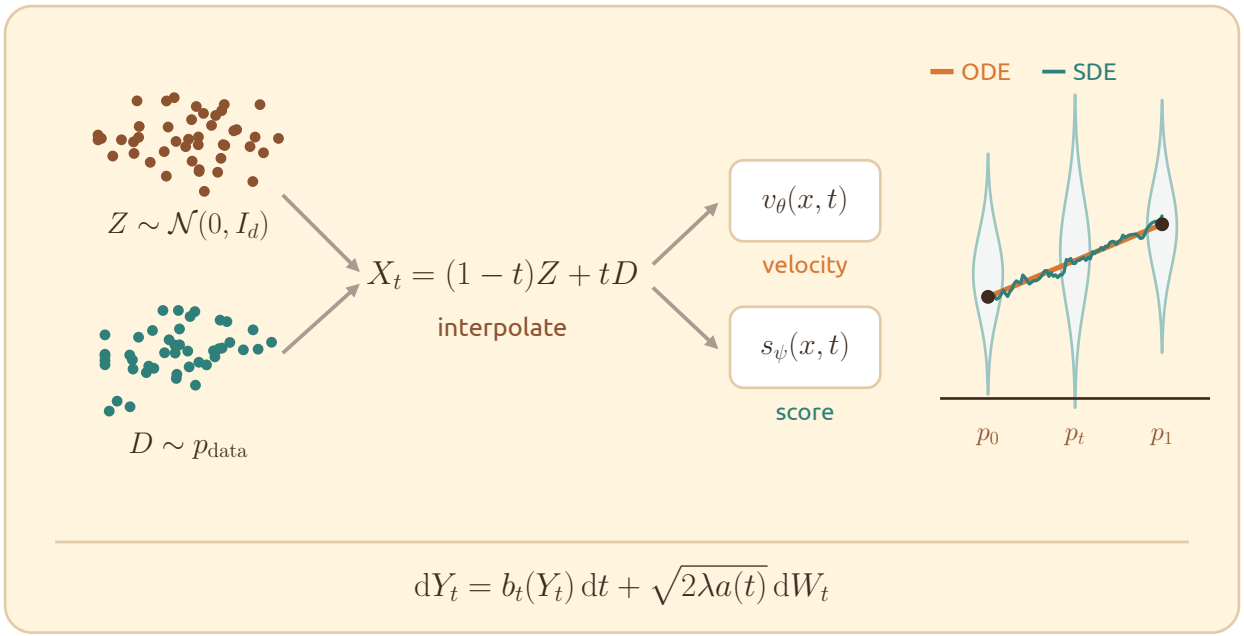


Figure 1. We will build intermediate states X_t from Gaussian noise Z and an RNA-FM microRNA embedding D, then fit a velocity v_θ and a score s_ψ with network parameters θ and ψ. The plot on the right follows one path under both samplers, with the shaded slices being the marginals p_t. In the sampling equation, b_t is the drift, W_t is Brownian motion, $\lambda \geq 0$ controls noise strength, and $a(t) > 0$ specifies its time dependence. You will derive b_t so that the sampler follows the target densities. The trajectories and density slices are schematic.

Research brief and notation

Throughout the main construction, generation time t increases from noise at $t = 0$ to data at $t = 1$. We can draw independent random vectors

$$Z \sim \mathcal{N}(0, I_d), \qquad D \sim p_{\text{data}}, \qquad X_t = (1 - t)Z + tD.$$

Here D is the embedding of a microRNA drawn from the training set, and d is the embedding width, 640 for the RNA-FM model your lab uses. The intermediate state X_t combines the Gaussian starting sample Z with the data sample D. The identity matrix I_d has d rows and columns. Write $\mu = \mathbb{E}[D]$ and $\Sigma = \text{Cov}(D)$.

Let p_t denote the marginal density of X_t, and let $q_t(x \mid D)$ denote its conditional density given the data sample. The target score is $s_t(x) = \nabla_x \log p_t(x)$. The target velocity v_t transports p_t. A sampler's actual density will be written ρ_t.

All spatial derivatives act on x. Assume smooth positive densities, finite displayed integrals and required moments, vanishing boundary terms, and conditions that justify differentiation under integrals and uniqueness of the stated ODE, SDE, and density equations. Conditional expectations indexed by t hold that time fixed. Score-training questions use $0 < t < 1$. We will denote natural logarithms by log.

Step	**Points**
1. Build the probability path	2.0
2. Train the fields and recover diffusion	3.0
3. Prove the new sampler's properties	3.0
4. Construct and test a finite-interval update	3.0
5. Add conditional and multi-objective guidance	1.5
6. Interpret benchmarks and ablations	2.5
Total: 30 parts	**15.0**

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Reference Formulas

Probability and Gaussians

Mean and covariance of an affine map
$$\mathbb{E}[AX+b] = A\mu + b, \quad \mathrm{Cov} = A\Sigma A^\top$$
Gaussian noising, independent terms
$$X_t = \alpha_t X_0 + \sigma_t \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I)$$
$$\mathrm{Cov}(X_t) = \alpha_t^2 \,\mathrm{Cov}(X_0) + \sigma_t^2 I$$
Score of a Gaussian
$$\nabla_x \log \mathcal{N}(x; m, \sigma^2 I) = -\frac{x-m}{\sigma^2}$$
Bayes' rule
$$p(x \mid y) = \frac{p(y \mid x)\, p(x)}{p(y)}$$
Total probability and posterior weight
$$p(x) = \int p(x \mid y) p(y)\, dy, \;\; w(y \mid x) = \frac{p(x \mid y) p(y)}{p(x)}$$

Linear algebra

Diagonalization, matrix exponential and its determinant
$$A = V \Lambda V^{-1}, \qquad e^{tA} = V e^{t\Lambda} V^{-1}$$
$$\det e^{hM} = \textstyle\prod_i e^{h m_i} = e^{h\,\mathrm{tr} M}$$
Divergence is a Jacobian trace
$$\nabla \cdot v = \mathrm{tr}\, J_v = \sum_i \partial_i v_i$$
Laplacian
$$\nabla^2 f = \nabla \cdot \nabla f = \sum_i \partial_i^2 f$$

Change of variables

Pushforward
$$\mathbb{P}(Y \in B) = \mathbb{P}\big(X \in \phi^{-1}(B)\big)$$
Density under an invertible map
$$p_Y(\phi(x)) = \frac{p_X(x)}{|\det J_\phi(x)|}$$
Log-density form
$$\log p_Y(\phi(x)) = \log p_X(x) - \log |\det J_\phi(x)|$$

Flows and densities

Trajectory and flow map
$$\frac{dx_t}{dt} = v_t(x_t), \qquad x_t = \phi_t(x_0)$$
Euler step
$$\hat{x}_{k+1} = \hat{x}_k + \Delta t\, v_{t_k}(\hat{x}_k)$$
Continuity equation and flux
$$\partial_t p_t + \nabla \cdot (p_t v_t) = 0, \quad j_t = p_t v_t$$
Score identity
$$s_t = \nabla \log p_t \iff p_t\, s_t = \nabla p_t$$
Log density along a trajectory
$$\tfrac{d}{dt} \log p_t(x_t) = -\nabla \cdot v_t(x_t)$$
$$\log p_1(x_1) = \log p_0(x_0) - \int_0^1 \nabla \cdot v_t \, dt$$

Flow matching

Linear interpolant and target
$$X_t = (1-t)X_0 + tX_1, \qquad U = X_1 - X_0$$
General schedule
$$X_t = \alpha_t X_0 + \beta_t X_1, \;\; U = \dot{\alpha}_t X_0 + \dot{\beta}_t X_1$$
Marginal velocity
$$v_t(x) = \mathbb{E}[\, U_t \mid X_t = x \,]$$
Least squares gives the conditional mean
$$\arg\min_a \mathbb{E}\big[\|a - U\|^2 \mid X_t = x\big] = \mathbb{E}[U \mid X_t = x]$$
Conditional flow-matching equivalence
$$\mathcal{L}_{\mathrm{CFM}} = \mathcal{L}_{\mathrm{FM}} + \mathrm{const}, \;\; \nabla_\theta \mathcal{L}_{\mathrm{CFM}} = \nabla_\theta \mathcal{L}_{\mathrm{FM}}$$

Stochastic dynamics

SDE and Euler–Maruyama step
$$dX_t = f(X_t, t)\, dt + g(t)\, dW_t$$
$$x_{k+1} = x_k + fh + g\sqrt{h}\, \varepsilon_k$$
Itô's lemma
$$d\varphi(X_t) = \Big[b \cdot \nabla\varphi + \tfrac{1}{2}\mathrm{tr}\big(\sigma\sigma^\top \nabla^2 \varphi\big)\Big] dt + \nabla\varphi^\top \sigma\, dW_t$$
Fokker–Planck equation
$$\partial_t p_t = -\nabla \cdot (f\, p_t) + \tfrac{1}{2} g^2 \nabla^2 p_t$$
Heat equation
$$\partial_t p_t = D\, \nabla^2 p_t$$
Ornstein–Uhlenbeck process
$$dX_t = -\kappa X_t\, dt + g\, dW_t$$
$$X_t \mid x_0 \sim \mathcal{N}\Big(e^{-\kappa t} x_0,\; \tfrac{g^2}{2\kappa}\big(1 - e^{-2\kappa t}\big) I\Big)$$

Diffusion

Variance-preserving corruption
$$X_r = \alpha_r X_0 + \sigma_r \varepsilon, \qquad \alpha_r^2 + \sigma_r^2 = 1$$
Score, noise and clean estimate
$$s = -\frac{\varepsilon_\theta}{\sigma_r}, \qquad \hat{x}_0 = \frac{x + \sigma_r^2 s}{\alpha_r}$$
Score matching by parts
$$J(\psi) = \mathbb{E}_p\big[\tfrac{1}{2}\|s_\psi\|^2 + \nabla \cdot s_\psi\big] + C$$
Reverse-time SDE
$$dX_t = \big[f - g^2 s_t\big] dt + g\, dW_t, \quad dt < 0$$
Probability-flow ODE
$$\frac{dX_t}{dt} = f(X_t, t) - \tfrac{1}{2} g(t)^2 s_t(X_t)$$
DDPM forward chain
$$x_k = \sqrt{\bar{\alpha}_k}\, x_0 + \sqrt{1 - \bar{\alpha}_k}\, \varepsilon, \;\; \bar{\alpha}_k = \prod_{i \le k} \alpha_i$$
DDPM reverse posterior
$$\tilde{\mu}_k = \frac{\sqrt{\bar{\alpha}_{k-1}}\, \beta_k\, x_0 + \sqrt{\alpha_k}\,(1 - \bar{\alpha}_{k-1})\, x_k}{1 - \bar{\alpha}_k}$$
$$\tilde{\beta}_k = \frac{\beta_k (1 - \bar{\alpha}_{k-1})}{1 - \bar{\alpha}_k}$$

Guidance

Conditional score
$$\nabla \log p(x \mid c) = \nabla \log p(x) + \nabla \log p(c \mid x)$$
Guidance scale and tilted target
$$s_t(x) + \gamma \nabla \log p_t(c \mid x)$$
$$\iff p_t^\gamma(x) \propto p_t(x) \cdot \big[p_t(c \mid x)\big]^\gamma$$
Reward guidance
$$s^{(R)} = s + \eta \nabla R, \qquad p^{(R)} \propto p\, e^{\eta R}$$
Where the guidance term enters
$$\text{SDE drift}: \; f - g^2\big(s + \gamma \nabla \log \ell\big)$$
$$\text{ODE velocity}: \; f - \tfrac{1}{2} g^2 \big(s + \gamma \nabla \log \ell\big)$$
Classifier-free guidance
$$F_{\mathrm{CFG}} = F_\varnothing + w\,(F_c - F_\varnothing)$$
Multi-objective weighting
$$\tilde{R}_m = \frac{R_m - \mu_m}{\tau_m}, \qquad R_\lambda = \sum_m \lambda_m \tilde{R}_m$$

Information measures

Entropy and differential entropy
$$H(p) = -\int p \log p, \qquad H(p, q) = -\int p \log q$$
KL divergence and its decomposition
$$D_{\mathrm{KL}}(p \| q) = \mathbb{E}_p\Big[\log \tfrac{p}{q}\Big] \;\ge 0$$
$$H(p, q) = H(p) + D_{\mathrm{KL}}(p \| q)$$

Notation Reference

Times and clocks

Symbol	Meaning
t	Generation time: noise at $t = 0$, data at $t = 1$
r	Noising clock: clean data at $r = 0$, noise at $r = T$
$\tau = T - r$	Generation clock for the diffusion model
s, t	Two times of one interval, $s < t$
h	Step size of a numerical update

Context-dependent symbols

Symbol	Meaning
τ_1, τ_2	Positive reward scales. *Not* a time
c	Strength of the score correction, a real constant
$c_{s,t}$	Noise coefficient of the interval update
c (as a condition)	Class label, as in p_t^c and ℓ_r^c
α_j, $\bar{\alpha}_j$	DDPM schedule and its running product
α	Weight between the two objectives, $0 < \alpha < 1$
μ, Σ	Mean and covariance of the data sample D
$\widetilde{\mu}_k$	DDPM reverse posterior mean
μ_1, μ_2	Reference means of the two rewards

Operators

Symbol	Meaning
∇	Gradient in x
$\nabla\cdot$	Divergence in x, equal to tr of the Jacobian
Δ, ∇^2	Laplacian, $\nabla\cdot\nabla$. The two are the same here
$\|\cdot\|$	Euclidean norm

Densities and fields

Symbol	Meaning
p_t, $q_t(\cdot \mid D)$	Target marginal; conditional given the endpoint
ρ_t	The density the sampler actually has
π_t	Guided density
v_t, s_t, b_t	Target velocity, target score, sampler drift
v_θ, s_ψ	The learned velocity and score
U_t, G_t	Conditional velocity and score targets
λ, $a(t)$	Noise strength and its time dependence
w, η	Classifier-free and reward guidance strengths
κ, ν	Ablation switches, each 0 or 1
ζ	Unit-rescaling factor, $\zeta > 1$

Step 1: Building the Probability Path (2 points)

Before designing a sampling rule, we need to specify the distribution of latents at each intermediate time. We'll use the interpolant in the research brief to derive training targets and examine how sample motion changes density, and then reuse these quantities when we construct and validate the sampler.

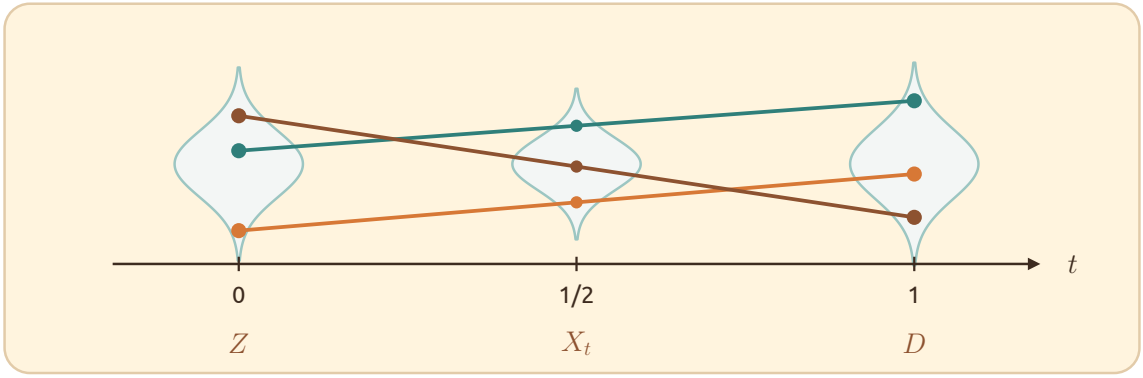


Figure 2. Each sampled pair of endpoints gives one straight path in latent space, and X_t is the state on that path at time t. A dot marks that state at $t = 0$, $t = 1/2$ and $t = 1$, and the shaded slices are the marginals p_t.

(a) Before training, we need the spread of the latents at each intermediate time. That spread becomes our reference later, when we check whether a sampling rule reproduces the intended distribution.

For the independent endpoints in the research brief, first derive the covariance of X_t. Box one equation for $\mathrm{Cov}(X_t)$ in terms of t, I_d, and Σ. (0.5 points)

(b) We now know the spread of the interpolated states. However, to move an individual latent along this path, we need a velocity target for each sampled pair of endpoints.

If we hold the sampled endpoints Z and D fixed and differentiate the interpolation, what conditional velocity target $U_t = \frac{\mathrm{d}}{\mathrm{d}t} X_t$ should the flow network regress against? (0.5 points)

(c) We also need a score network to estimate the direction in which log density increases. During training we know which microRNA the noise was added to, and conditioning on that endpoint leaves a Gaussian we can differentiate.

Derive the conditional score $\nabla_x \log q_t(x \mid D)$ in terms of x, D, and t. (0.5 points)

(d) As latents move, local expansion or contraction changes their density. Before we prove conservation for the stochastic sampler, let's work through a solvable linear flow that connects sample motion, the Jacobian of its map, and the resulting density change.

Consider $\dot{y} = My$ with known diagonalization $M = P\,\mathrm{diag}(m_1, \ldots, m_d)P^{-1}$, where P is invertible and $m_i \in \mathbb{R}$. Let r_0 and r_h be the densities before and after time $h > 0$, and let Φ_h be the flow map. Derive the map and its Jacobian, then box only the density ratio $r_h(\Phi_h(y))/r_0(y)$. (0.5 points)

Step 2: Training the Fields and Recovering Diffusion (3 points)

During training, we draw a complete endpoint pair and evaluate the conditional targets from Step 1. During generation, however, we have only the current state and time, so the networks must provide the velocity and score from those alone. Let's now justify training them on the sampled targets.

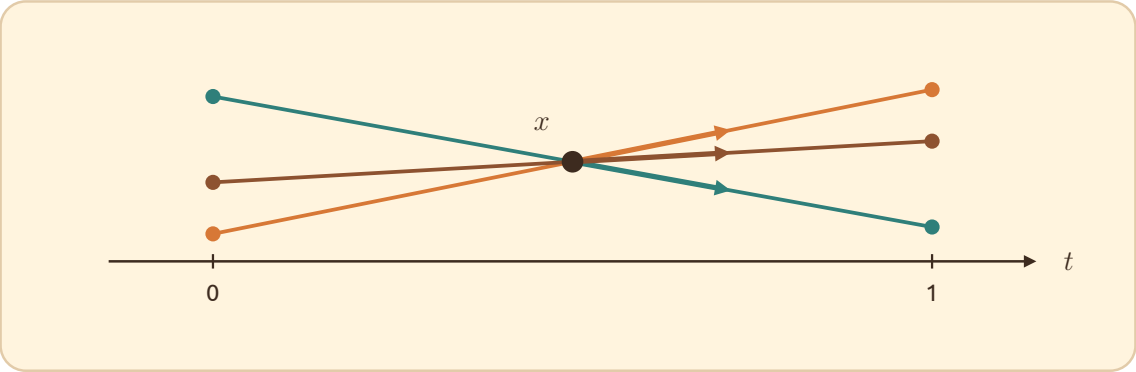


Figure 3. Three sampled pairs pass through the same latent x, and the short arrows are their conditional velocities $U_t = D - Z$. The network sees only (x, t), so it must return one field there.

Let $G_t = \nabla_x \log q_t(X_t \mid D)$. If we draw t uniformly from $(0, 1)$ and train networks with parameters θ and ψ using

$$\mathcal{L}_v(\theta) = \mathbb{E}\|v_\theta(X_t, t) - U_t\|^2, \qquad \mathcal{L}_s(\psi) = \mathbb{E}[(1-t)^2\|s_\psi(X_t, t) - G_t\|^2].$$

we can fit the velocity and score to the conditional targets from Step 1. Both networks receive only (X_t, t).

(a) Many microRNAs can produce the same intermediate latent, and each one arrives there with its own velocity. At generation time we observe only the current state and time, so the

network has to return a single velocity. We need to establish what it learns when we train it against these competing targets.

Use a squared-error decomposition to characterize the population minimizer of $\mathcal{L}_v$. Box one equation for the optimal marginal velocity $v_t(x)$ as a conditional expectation. Use the derivation to justify regressing against the sampled conditional velocity. Only the boxed conditional expectation will be graded. (0.5 points)

(b) We can apply the same conditional-to-marginal reasoning to the score. At an observed latent, some clean microRNAs are more plausible than others, and we need a weighting of their conditional scores that targets the marginal distribution we sample from.

The marginal density is $p_t(x) = \int q_t(x \mid y) p_{\text{data}}(y)\, \mathrm{d}y$. Here y denotes a possible data endpoint. We can differentiate the density to express its score as a posterior-weighted average:

$$s_t(x) = \int w_t(y \mid x)\, \nabla_x \log q_t(x \mid y)\, \mathrm{d}y.$$

Which normalized weight $w_t(y \mid x)$ satisfies this identity? (0.5 points)

(c) The score describes local density change. We also want the clean latent underneath a noisy one, because the language-model head decodes that latent into nucleotides. The posterior averaging identity from (b) will connect the two predictions.

Derive $\widehat{D}_\psi(x, t)$ using only x, t, and $s_\psi(x, t)$. At an exact score, this estimator should equal $\mathbb{E}[D \mid X_t = x]$. (0.5 points)

(d) Our score targets come from known perturbations of real microRNAs, while the original score-matching objective measures error against the unknown marginal score. If we rewrite that loss, we can train from samples alone.

At a fixed time, the ideal score loss is

$$J(\psi) = \frac{1}{2}\mathbb{E}_{X\sim p}\|s_\psi(X) - \nabla \log p(X)\|^2.$$

Integrate the cross term by parts. Write the complete integrand $F_\psi(x)$ in the equivalent objective $J(\psi) = \mathbb{E}_p[F_\psi(X)] + C_p$, where C_p is independent of ψ. (0.5 points)

(e) We have the learned score and the clean estimate, so now let's look at how denoising produces the next state. In a DDPM reverse step, both the current noisy latent and the clean endpoint enter the posterior mean. We should derive that mean with the known endpoint before we consider a learned replacement.

Assume the clean state x_0 is known. Let $k \geq 2$, $0 < \beta_j < 1$, $\alpha_j = 1 - \beta_j$, and $\bar{\alpha}_j = \prod_{i=1}^{j} \alpha_i$. The forward densities are

$$X_{k-1} \mid x_0 \sim \mathcal{N}(\sqrt{\bar{\alpha}_{k-1}} x_0, (1 - \bar{\alpha}_{k-1}) I_d),$$
$$X_k \mid X_{k-1} = u \sim \mathcal{N}(\sqrt{\alpha_k} u, \beta_k I_d).$$

Complete the square in their product. Box one equation for the posterior mean $\widetilde{\mu}_k(x_k, x_0)$ of $q(x_{k-1} \mid x_k, x_0)$. (0.5 points)

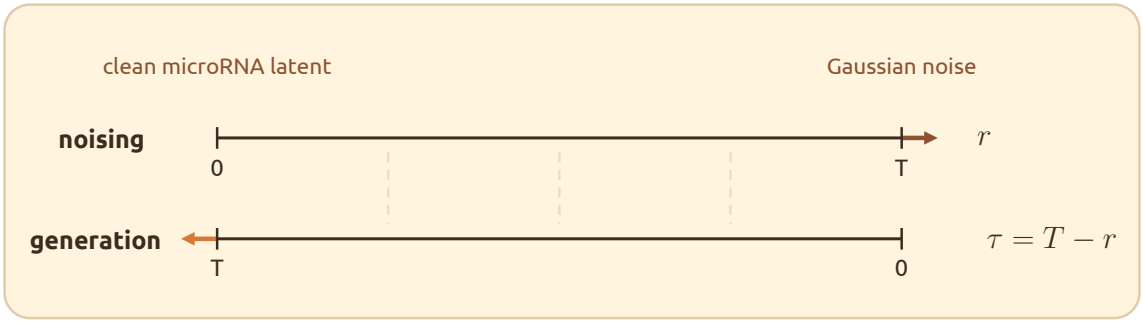


Figure 4. The noising clock r runs from a clean microRNA latent to Gaussian noise, and generation runs the other way on $\tau = T - r$. The dashed lines join the states that carry both time labels, and a drift written in r changes sign in τ.

(f) The DDPM posterior gives us one discrete denoising step. However, our sampler is continuous, so we need deterministic and stochastic reverse dynamics. Because generation traverses the noising process in the opposite direction, we need to keep track of time (Figure 4).

Consider a forward noising process with clock r:

$$\mathrm{d}X_r = f_r(X_r)\mathrm{d}r + g_r\mathrm{d}W_r.$$

Generation uses the increasing clock $\tau = T - r$. At the same state and corresponding noising time, which pair gives the probability-flow ODE velocity and reverse-SDE drift? Here $s = \nabla \log p_r$, and f and g also refer to noising time r. Choose ONE.

1. ODE: $f - \frac{1}{2}g^2 s$; SDE: $f - g^2 s$.
2. ODE: $-f + g^2 s$; SDE: $-f + \frac{1}{2}g^2 s$.
3. ODE: $-f + \frac{1}{2}g^2 s$; SDE: $-f + g^2 s$.
4. ODE: $-f - \frac{1}{2}g^2 s$; SDE: $-f - g^2 s$.

(0.5 points)

Step 3: Proving the Sampler's Properties (3 points)

You can now learn a velocity and a score from sampled training pairs. The proposed sampler combines them with a tunable amount of Brownian noise. To establish that it is valid, derive the density evolution and require agreement with the probability path chosen at the start.

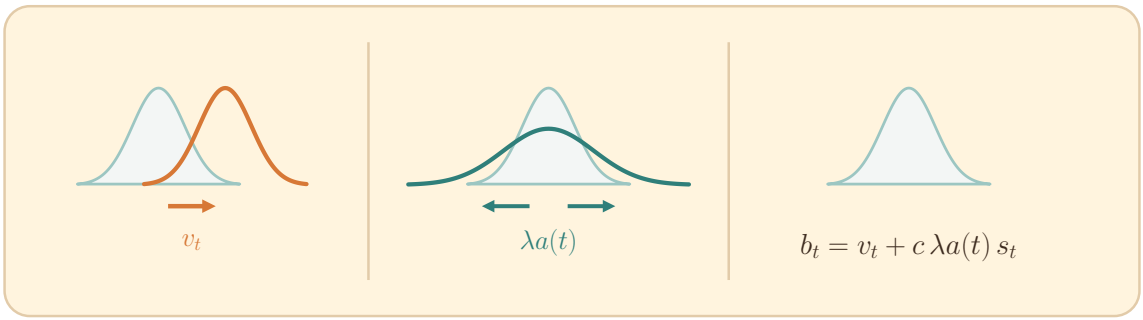


Figure 5. On the left the velocity v_t transports the density, and in the middle the Brownian term of strength $\lambda a(t)$ spreads it. The drift on the right carries a score correction of unknown strength c, which you will determine in part (c).

For the proof, assume the velocity and score are exact and satisfy

$$\partial_t p_t = -\nabla \cdot (p_t v_t), \qquad s_t = \nabla \log p_t.$$

Let $\lambda \geq 0$ be a constant noise-strength parameter, and let $a(t) > 0$ be a known scalar function of time. Consider the Itô SDE

$$\mathrm{d}Y_t = b_t(Y_t)\mathrm{d}t + \sqrt{2\lambda a(t)}\,\mathrm{d}W_t,$$

where W_t is standard d-dimensional Brownian motion. Derive a drift b_t that preserves the chosen marginals p_t.

(a) We will begin the density proof by following a smooth measurement of a random latent. Itô's lemma describes how its average changes under drift and Brownian fluctuations. Let's use that relation to move from individual trajectories to the evolution of the full distribution.

Let $\varphi : \mathbb{R}^d \to \mathbb{R}$ be a smooth, time-independent test function. Apply Itô's lemma and take expectations. Write $\mathcal{A}_t\varphi(x)$ in

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}[\varphi(Y_t)] = \mathbb{E}[\mathcal{A}_t\varphi(Y_t)].$$

The operator must include both drift and curvature terms. (0.5 points)

(b) The identity in (a) describes changes in averages of test functions. To compare the proposed sampler with the target path, we need an equation for the density itself.

Use integration by parts in the weak identity from (a) to obtain the Fokker–Planck equation for the sampler density ρ_t. Write its complete right-hand side. (0.5 points)

(c) We can now test whether the intended density satisfies the sampler's evolution equation. Adding noise spreads probability, so the drift has to compensate for us to preserve the chosen path. As such, we need to match the two density equations to determine the strength of the score correction.

We can restrict the drift to the family

$$b_t = v_t + c\,\lambda a(t) s_t,$$

where c is a real constant. Derive the value of c that makes $\rho_t = p_t$ solve the sampler's density equation for every admissible p_t, v_t, $a(t)$, and $\lambda > 0$. Give the complete corrected drift $b_t(x)$. (0.5 points)

(d) With the target evolution matched, we should check that the dynamics keep total probability normalized. Writing the density equation in terms of probability flux connects the local calculation in (b) to a conservation statement for the whole distribution.

Combine the drift and diffusion fluxes into a probability current J_t so that the sampler's density equation has the conservation form $\partial_t \rho_t = -\nabla \cdot J_t$. Integrate this equation using the stated boundary assumptions to check conservation of total probability. Box only the equation for J_t. (0.5 points)

(e) We have a compatible density equation and we have checked mass conservation. To complete the main claim, we need to connect these calculations to the law of the sampler we run from noise. We'll use the initial condition and uniqueness to determine what holds over the entire time interval.

Using the corrected drift, the common initial density $\rho_0 = p_0$, and the uniqueness assumptions, which statement completes the marginal-agreement theorem? Choose ONE.

1. Conservation of total mass alone proves $\rho_t = p_t$ for any drift.
2. For every $\lambda \geq 0$, the sampler has $\rho_t = p_t$. At $\lambda = 0$ it follows the flow ODE, and positive λ generally changes individual trajectories.
3. All values of λ give identical trajectories when the initial sample is fixed.
4. Agreement holds only at $t = 1$, with no requirement on intermediate densities.

(0.5 points)

(f) The density argument also gives us a likelihood for every microRNA the ODE generates – we can report this likelihood when we rank designs. Along a trajectory, local expansion lowers density and contraction raises it. Let's derive the accumulated change in log density.

Combine the continuity equation with the multivariable chain rule to derive the change in log density along $\dot{Y}_t = v_t(Y_t)$. Integrate from 0 to 1. Box one equation for $\log p_1(Y_1)$ in terms of the initial log density and an integral along the trajectory. (0.5 points)

Step 4: Constructing a Finite-Interval Update (3 points)

The continuous dynamics now have the density evolution we wanted. Our implementation will advance via finite intervals, so we need to know which updates preserve that behavior. If we take one coordinate of the embedding and give it a Gaussian path, the interval update comes out in closed form.

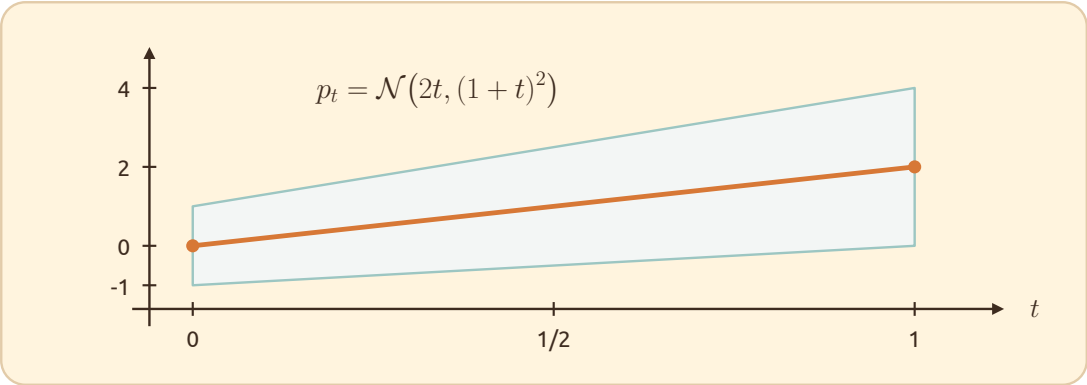


Figure 6. The path used throughout Step 4. The line is the mean, which runs from 0 to 2, and the band covers one standard deviation on either side, which grows from 1 to 2.

For this step, let's choose coupled endpoints $D = 2 + 2Z$ with $Z \sim \mathcal{N}(0, 1)$, giving

$$X_t = 2t + (1+t)Z, \qquad p_t = \mathcal{N}(2t, (1+t)^2).$$

The mean moves from zero to two, while the standard deviation grows from one to two. We'll compute the marginal fields directly from this coupled path.

(a) Both the mean and the spread change with time here. To implement the ODE we need a velocity at the current position that accounts for both motions. Let's convert the known path into a field that depends only on the observed state and time.

First, differentiate the path while holding Z fixed, then eliminate Z using the current position x. What deterministic velocity $v_t(x)$ generates this Gaussian path? (0.5 points)

(b) With the instantaneous velocity in hand, we want to advance across an entire interval in one update. An exact flow map lets us check a numerical ODE solver and compare deterministic motion with the stochastic interval update we develop next.

Derive the flow map $\Phi_{s,t}(x)$ that takes state x at time s to time t, with $0 \le s < t \le 1$. Use the same value of Z at both times. Box one equation for $\Phi_{s,t}(x)$. (0.5 points)

(c) To add adjustable stochasticity to this example, we separate the changing mean and spread from the remaining random fluctuations. Standardizing the state removes the known translation and scale. We can then examine the effect of the noise-strength parameter against a fixed reference distribution.

Use the corrected SDE from Step 3 with $a(t) = (1 + t)^2$. Define the standardized state $H_t = (Y_t - 2t)/(1 + t)$. Apply Itô's lemma to derive the SDE for H_t. (0.5 points)

The standardized process has a linear restoring drift. We can use its conditional-mean contraction $r_{s,t} = e^{-\lambda(t-s)}$. Let $\xi \sim \mathcal{N}(0,1)$ be independent of the state at time s.

(d) In standardized coordinates, let's split a future state into the part explained by the current state and an independent Gaussian draw. Over a finite interval, we can choose the noise amplitude to balance the contraction of the conditional mean and use this exact update to assess the approximations in the next two parts.

Let's construct an exact affine Gaussian update of the form

$$Y_t = 2t + (1+t)\left[r_{s,t}\frac{Y_s - 2s}{1+s} + c_{s,t}\xi\right], \qquad c_{s,t} \geq 0.$$

We need to determine $c_{s,t}$ so that the standardized variance stays equal to one whenever it starts at one. Write the complete update, including your noise coefficient. (0.5 points)

(e) We could turn the stochastic interval update into a faster deterministic rule by keeping only its conditional mean. Before we rely on that shortcut, let's first check whether it preserves the spread of the target distribution.

We can replace the exact stochastic update by $\widetilde{Y}_t = \mathbb{E}[Y_t \mid Y_s]$, with $Y_s \sim p_s$ and $s < t$. Derive the fraction of the target variance retained by this shortcut. Box one expression for $\mathrm{Var}(\widetilde{Y}_t)/(1+t)^2$ in terms of λ and $t - s$. (0.5 points)

(f) A second approximation keeps the stochastic dynamics and takes Euler-Maruyama steps. Errors in each step can accumulate through repeated updates, so we should find the variance that the numerical recursion settles on.

We can compare this shortcut with repeated Euler-Maruyama steps for the standardized process from (c). We'll use a fixed step size h with $0 < \lambda h < 2$ and independent standard Gaussian increments. Let V_h be the invariant variance of this numerical chain, meaning that its variance is unchanged after one step. Derive the variance recursion and box one equation for V_h. (0.5 points)

Step 5: Conditional and Multi-Objective Guidance (1.5 points)

The sampler is unconditional and you have a finite-step reference for it. A useful microRNA has to silence the transcript you are targeting and leave other transcripts untouched, so generation must be steered. You will extend the score through conditional probabilities and reward weighting, then check whether the dynamics follow the resulting density path.

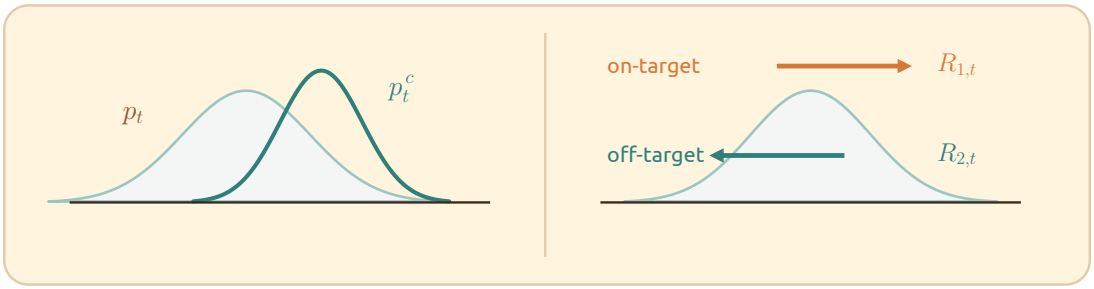


Figure 7. On the left, a class label tilts the density p_t toward the microRNAs that act on our chosen transcript, giving p_t^c. On the right, the reward $R_{1,t}$ pulls the sample toward stronger on-target silencing and $R_{2,t}$ pulls it toward weaker off-target silencing. The two are reported in different units.

(a) So far generation has been unconditional. To draw only microRNAs that act on our chosen transcript, we need to account for how the probability of that class varies across noisy latents. First, we should connect a classifier's output to the conditional density, then determine how the resulting score changes the reverse dynamics.

Begin with classifier guidance for the diffusion model from Step 2(f). In forward noising time r, let $s_r = \nabla \log p_r$ and let $\ell_r^c(x) = \Pr(c \mid X_r = x) > 0$ for a class with positive prior probability. The noising coefficients f_r and g_r are independent of the class. Use Bayes' rule to derive the conditional score, then reverse the clock using $\tau = T - r$. Box one equation for the class-conditional reverse-SDE drift $b_\tau^c(x)$. Use guidance strength of 1 and exact fields. (0.5 points)

(b) The class label still leaves wide variation in on-target silencing and in off-target silencing. We want to extend the guidance calculation to both, with separate controls for their relative importance and the overall strength. One is a predicted knockdown and the other counts affected transcripts, so we have to handle their units. Let's obtain the class contribution from the conditional and unconditional scores.

Return to generation time t. Let $p_t^c(x) = p_t(x \mid c) > 0$, and suppose the unconditional and conditional scores are exact. We will use classifier-free strength w, with $w = 0$ unconditional and $w = 1$ conditional.

But we also want to maximize property $R_{1,t}(x)$ and minimize property $R_{2,t}(x)$. We can normalize each property with a fixed reference mean μ_i and positive scale τ_i, and assign relative weights α and $1 - \alpha$, where $0 < \alpha < 1$, and apply reward-guidance strength $\eta \geq 0$ after classifier-free guidance.

Derive the density whose score equals this combined guided score. Assume it is normalizable. Box one equation for $\pi_t(x)$, including both raw rewards and their normalization factors. You may denote its normalizing constant by Z_t. (0.5 points)

(c) We now have a density for the guided score. However, we should check whether the sampler follows it as time advances. That density has to be compatible with the velocity we kept from the unconditional construction. So we should compute the continuity residual to measure any disagreement between the two.

Let's write the density from (b) as

$$\pi_t(x) = \frac{p_t(x)e^{\chi_t(x)}}{Z_t}, \qquad \chi_t = w \log \frac{p_t^c}{p_t} + \eta R_t^{(\alpha)},$$

where $R_t^{(\alpha)}$ is the signed, normalized multi-objective reward constructed in (b), and w, η, α are constant. Replace the score in the corrected sampler by $\nabla \log \pi_t$, retain v_t, and initialize from π_0.

Use the continuity equation for p_t and the product rule to compute the normalized continuity residual

$$\mathcal{C}_t(x) = \frac{\partial_t \pi_t(x) + \nabla \cdot (\pi_t(x) v_t(x))}{\pi_t(x)}.$$

Box one expression for $\mathcal{C}_t$ using only χ_t, v_t, and Z_t. The guided path is valid with the retained velocity exactly when this residual vanishes. (0.5 points)

Step 6: Interpreting Benchmarks and Ablations (2.5 points)

To support a methods paper, we must compare the sampler with a flow ODE and use controlled ablations to examine why performance changes. The distribution error is the terminal $D_{\mathrm{KL}}(\rho_1 \| p_1)$, where lower is better.

The velocity and score networks have equal cost per call. The ODE evaluates only the velocity network once per step. The corrected SDE evaluates both networks once each per step.

Run	Sampler	Steps	Distribution error
A	ODE	20	0.12
B	ODE	40	0.06
C	Corrected SDE	20	0.04
D	Corrected SDE	40	0.025

(a) Before we make a performance claim, we need to compare the samplers under the same computational budget. One stochastic step requires both neural fields, so a step count alone does not measure cost.

Which benchmark conclusion compares the samplers at the same number of network calls? Choose ONE.

1. Compare A with C because both use 20 steps, and conclude that the compute budgets are equal.
2. Compare B with C. Both use 40 network calls, and C has lower error in this benchmark.
3. Compare B with D because both use 40 steps, and conclude that the compute budgets are equal.
4. Select D's lowest error and conclude that stochastic sampling is always more accurate at equal cost.

(0.5 points)

To separate field error from integration error, vary step size while keeping data, evaluation, stochasticity, and sampler fixed. Compare exact analytic fields with learned fields.

Step size	**Exact-field error**	**Learned-field error**
h	0.080	0.120
$h/2$	0.020	0.060
$h/4$	0.005	0.045
$h/8$	0.00125	0.04125

(b) A decrease in error with smaller steps can reflect more accurate numerical integration. To decide what we improve next, use the exact-field runs as a control and examine how both columns change with step size. Select the follow-up that addresses the limitation this comparison exposes.

Which next action most directly addresses the limiting error suggested by this ablation? Choose ONE.

1. Increase the number of steps indefinitely and keep both learned fields fixed.
2. Remove the score correction while keeping all injected noise.

3. Check and improve the learned velocity and score, then repeat the step-size study at matched computation.

4. Conclude that probability conservation guarantees the learned fields are correct.

(0.5 points)

(c) To explain the step-size study in the paper, we need to connect network inaccuracies to the density evolution. Substitute the learned fields into the sampler equation and identify the term that survives the exact identities. That term shows how field errors affect the continuous dynamics.

Write the learned fields as $v_\theta = v_t + e_v$ and $s_\psi = s_t + e_s$, where e_v and e_s are field errors. Evaluate the learned sampler's density defect at the target density:

$$\mathcal{E}_t^\lambda = -\nabla \cdot \left[p_t(v_\theta + \lambda a(t) s_\psi)\right] + \lambda a(t) \Delta p_t - \partial_t p_t.$$

Use the exact continuity and score identities to simplify it. Box one expression for $\mathcal{E}_t^\lambda$ in terms of p_t, e_v, e_s, λ, and $a(t)$. (0.5 points)

(d) We should also check whether the multi-objective comparison depends on the units a property is measured in. Reporting off-target silencing as a fraction of the transcriptome and reporting it as a percentage describe the same preference, yet one of them can change the influence of a raw reward. Let's compare the original and rescaled runs within each method.

We need to test the multi-objective normalization in Step 5(b) through a controlled change of units. Replace $R_{2,t}$, μ_2, and τ_2 by $\zeta R_{2,t}$, $\zeta\mu_2$, and $\zeta\tau_2$, with $\zeta > 1$. We'll keep the velocity and score networks, random seeds, guidance strengths, objective weights, and sampling budgets fixed, and the ablation will use the raw reward $\alpha R_{1,t} - (1-\alpha)R_{2,t}$.

The hypothetical held-out quality scores below are defined so that higher is better for both properties.

Reward	**Property-2 units**	Q_1	Q_2
Normalized	Original	0.72	0.68
Normalized	Rescaled	0.72	0.68
Raw	Original	0.70	0.63
Raw	Rescaled	0.41	0.88

For each reward construction, define the score change as $\Delta s = s_{\text{rescaled}} - s_{\text{original}}$. Which predicted score-change pair $(\Delta s_{\text{normalized}}, \Delta s_{\text{raw}})$ explains the distinction tested by this ablation? Choose ONE.

1. $\big(0,\ -\eta(1-\alpha)(\zeta-1)\nabla R_{2,t}\big)$.
2. $\big(-\eta(1-\alpha)(\zeta-1)\nabla R_{2,t},\ 0\big)$.
3. $(0, 0)$ because the weights are fixed.
4. $\big(0,\ +\eta(1-\alpha)(\zeta-1)\nabla R_{2,t}\big)$.

(0.5 points)

(e) Having tested the reward construction, we turn to the two components we added to the sampler itself. To interpret the removal of either the score correction or the noise, we need to work out how that change affects the density equation at the intended target.

With exact fields, consider

$$\mathrm{d}Y_t = \big(v_t(Y_t) + \kappa\lambda a(t) s_t(Y_t)\big)\mathrm{d}t + \sqrt{2\nu\lambda a(t)}\,\mathrm{d}W_t,$$

where $\kappa, \nu \in \{0, 1\}$ independently switch the score correction and noise on or off. Use the same initial density, $\lambda > 0$, and time-only $a(t) > 0$ for all variants. Define

$$\mathcal{B}_t^{\kappa,\nu} = \partial_t \rho_t|_{\rho_t = p_t} - \partial_t p_t.$$

Box one formula for this defect in terms of κ, ν, λ, $a(t)$, and p_t. It should distinguish the full sampler, either single-component ablation, and the flow ODE. (0.5 points)

End of exam.

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